

Recurrent world model (RSSM)  $h_t$

probe readout  $v^T h_t$

**1** Emergence

$C_t$ : accumulated  
predictive difficulty

**2** External validity  
beyond KL/Recon/EMA

future imagination error  
 $E_{t,K}^{\text{state}}$

**4** Intervention on dense  $v$

Internal readout

✓ strong

instantaneous signals:  
KL, Recon, EMAREcon

**3** Mechanism

dense  $v$ : diffuse

sparse atoms:  
#612, #156

sparse atoms ✓

External quality

✗ null (dense  $v$ )